BSc (Hons) Computer Science

University of Bedfordshire

Reflective Report

ElderConnect: Developing a Social Engagement and Wellness Platform Using Artificial Intelligence for Elderly People

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Module: Undergraduate Project (CIS017-3)

Degree: BSc (Hons) Computer Science

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1. Introduction

The final year undergraduate project is the one piece of work in the degree where you get to build something that actually matters to you, rather than following a brief someone else wrote. I did not take that lightly. The system I chose to develop, ElderConnect, came from a genuinely personal place - which both in terms of the problem it addresses and the kind of developer I wanted to prove I could be by the end of it.

ElderConnect is a cross-platform mobile application developed using Flutter, Supabase, and the Hugging Face Sentiment Analysis API. It is designed specifically for elderly users aged sixty and above, which combining social engagement, AI-powered mood detection, medication reminders, wellness activities, and a caretaker monitoring portal within a single, accessibility-first interface. The problem it addresses is social isolation and the absence of a genuinely usable digital platform for elderly people is one that existing solutions have failed to solve comprehensively, as my literature review and primary survey research confirmed early in the project.

This reflective report documents the journey from that initial concept to the current stage of development. It is not a summary of what I was built, but rather an honest account of how it was built - the planning decisions, the setbacks, the moments where I had to reconsider an approach, and the practices that genuinely helped. Writing it at this stage of the project, with the core development substantially underway and the final sprints in progress, has given me enough perspective to reflect critically rather than superficially.

This report is structured to cover my motivation and self-management, followed by a detailed reflection on development progress, challenges, and lessons learned throughout, concludes with a chapter-by-chapter outline of the final thesis.

2. Self-Management

2.1 Personal Motivation and Initiation

There is a particular kind of irony in the fact that the technologies meant to keep people connected are often the very ones that leave certain people behind. I did not arrive at this project through a research paper or a module brief. I arrived at it through watching someone I love become gradually excluded from the digital world she genuinely curious about.

My grandmother, by any measure, is a remarkably social person. She had a wide circle of friends and former colleagues that she spent decades building relationships with. She carried that sociability into later life with an enthusiasm that most people half her age would struggle to match even. What struck me the most was that the curiosity of her which never fades. She was genuinely interested in technology - as in the idea of it, in what it could do for her and in staying connected through it. She would ask questions, she would try things, and would persist longer than most people expected her to. But the applications available to her are simply not built with her in mind. The texts are too small, the navigation too layered, the logic too assumed. Each time she tried and found herself confused, it was not a failure of willingness on her part - it was a failure of design on the part of the developers who had never considered her as a user at all.

That observation stayed with me. When the time came to propose a final-year project, I did not need to search for a problem worth solving. I already knew what it was. The academic research I subsequently conducted during the literature review phase only reinforced what I had observed personally, that the social isolation among elderly individuals is a serious and well-documented health concern, and also the technology gap is a significant contributor to it. The survey data I collected from 52 elderly respondents and 13 caretakers confirmed the same picture in quantitative manner. Nearly 70% of elderly participants reported of limited confidence using smartphones, 73% experienced loneliness at least sometimes, and 84.6% of caretakers found it difficult to monitor the well-being of their elderly relatives. These were not abstract statistics, they mapped directly onto experiences I recognised.

Having both a personal reason and real survey data pointing in the same direction gave the project a clarity of purpose from the start that made it easier to stay focused, even during the more technically demanding phases.

2.2 Proactiveness and Risk Management

From the very beginning of this project, I made a conscious decision to stay ahead of problems rather than wait for them to arrive. That does sounds straightforward, but in practice it meant building risk thinking into every planning decision rather than treating it as a separate exercise to be done once and forgotten. When I reviewed the project landscape at the proposal stage, three risks stood out as genuinely which likely to cause problems if I did not address them early as possible. The complexity of integrating an external AI API, the challenge of designing for a user group I could not fully represent myself, and the risk of timeline slippage from underestimating accessibility work.

The Hugging Face Sentiment Analysis API integration was flagged from the outset as the most technically uncertain component of the system. Rather than leaving it as a final sprint task, I planned the AI integration as Sprint 3 and kept it separate from the caretaker portal in Sprint 5 specifically so that a problem in one would not block the other. This modular sprint design was a deliberate risk mitigation decision, not just an organisational preference. It meant that when Sprint 3 encountered edge case complexity around ambiguous sentiment results, the caretaker portal development could continue in parallel without stalling.

The accessibility risk was the one I was most concerned about because it was the hardest to anticipate precisely. I could not fully simulate the experience of an elderly user from behind a development screen. So rather than delaying user evaluation until a formal UAT phase at the project's conclusion, I integrated informal usability testing with target-demographic participants starting from Sprint 1. That proactive decision is what surfaced the tap target sizing issues and navigation depth problems early enough to fix them within the sprint rather than having to retrofit changes across a completed system. The printed questionnaire format for the elderly user survey was a similar call recognising before the data collection began that an online form would introduce a literacy barrier that would undermine the reliability of the responses, and switching format accordingly before it became a problem.

2.3 Time Management and Planning

One of the more honest things I can say about time management in a project of this scale is that the plan you create at the beginning is rarely the plan you end up following. And how well you manage that gap is what actually determines whether the project succeeds. I planned the ElderConnect project using a Gantt chart spanning from October 2025 to April 2026 which divided into seven phases as - Planning, Requirement Gathering and Research, Design, Development across six sprints, Testing, Evaluation, and Documentation. At the time of creating the plan, these phases looked cleanly sequential. In practice, the experience was considerably more iterative.

The planning and research phases completed largely on schedule, which gave me a solid foundation going into the design phase. However, it was during the design and early development stages that the timeline required adjustment. The design phase, which I had originally estimated at around 50% completion by the contextual report submission in late December 2025, proved to be more demanding than anticipated - not because of technical

difficulty, but because of something more nuanced. Designing an interface for elderly users is not as the same as designing any other interface. Every layout decision, every colour choice, every navigation structure had to be tested and reconsidered with the actual target user in mind. I hv conducted sevaral informal usability testing sessions with target-aged participants specifically to evaluate the UI and UX of the system, and the feedback from those sessions consistently surfaced some issues I had not anticipated from behind a development screen. This was time - consuming, but it was also one of the most valuable activities I undertook throughout this project.

As a result of these iterative UI and UX refinements, sprint 4, which covers health and wellness features has extended slightly beyond its originally planned dates. More significantly, rather than serving as a discrete, final endpoint as originally scoped, sprint 6, which focused on accessibility and refinement has effectively evolved into a continuous process integrated alongside every other sprint. Every time a feature was built, accessibility considerations required immediate attention rather than being deferred to a later polish pass. Recognising this and adjusting the plan accordingly, than forcing the project back into an unrealistic original structure, was a decision I am glad I made early.

Sprints 3 and 5, covering AI integration and the caretaker portal respectively, are currently in active development and debugging at the time of writing. These are the most technically complex components of the system which I have allocated the time accordingly in the revised plan. Offering a more comprehensive narrative of the project's evolution than the original plan alone, the updated Gantt chart provided in Appendix A highlights the variance between the initial schedule and the actual progression,.

To ensure and maintain accountability throughout, I have submitted monthly progress reports to my supervisor, Ms. Dilushinie Fernando, treating each review as a critical evaluative checkpoint rather than just a formality to get signed off. Those reports helped me catch when I was drifting from a milestone before it became a real problem. The feedback gathered during these meetings, particularly the recommendation to revise the initial prototype's color scheme - has directly influenced several development decisions, ultimately enhancing the final product. Having that consistent outside perspective running alongside my own judgment made a real difference, and I think without it I would have made slower progress and caught problems later than I did.

3. Reflection

3.1 Milestone Adherence and Current Project Status

Looking back at the Gantt chart I submitted with the contextual report and comparing it honestly against where the project stands today is a useful exercise, not because it reveals failure, but because the differences between the two tell a more accurate story of how software development actually works than any perfectly followed plan ever could.

The first two phases: Planning, and Requirement Gathering and Research, were completed on schedule and to a standard I am genuinely satisfied with. The decision to conduct primary research through two separate surveys, one distributed online to caretakers and one administered through printed questionnaires to elderly participants, was the right call both methodologically and practically. The printed format for elderly respondents was not something I had originally planned to use, it emerged from thinking carefully about who I was asking to participate and what barriers an online form would create for them. That small but deliberate decision produced richer, more reliable data than an online-only approach would have. It also reinforced something I would carry through the entire development process, not just their requirements on paper, also that designing for this user group requires a constant awareness of their context,.

By the time the contextual report was submitted in late December 2025, the design phase was approximately halfway complete, which aligned with the planned timeline. Sprints 1 and 2, covering environment setup and authentication followed by the social features, were completed broadly as planned. Sprint 4, which covered the health and wellness features including medication reminders, wellness games, and the emergency contact button, was also completed, though with the timeline extension discussed in the previous section. Sprint 6, which is constituted by the UI/UX refinement work, has been run continuously alongside the development sprints by me, rather than as a final standalone phase, and the accessibility testing sessions I conducted with target-age participants have fed directly and consistently into the build process throughout. Selected screenshots are provided in Appendix B, evidencing the current state of the implemented interface, demonstrating the accessibility-first design approach across key screens including the social feed, medication reminder system, and registration flow.

The table below summarises planned versus actual completion across the key project phases, providing a structured reference for the milestone adherence discussion in this section and the updated Gantt chart in Appendix A.

Table 1: Milestone Adherence

Phase / Milestone	Planned Completion	Actual Completion	Status
Planning	October 2025	October 2025	Completed ✓
Requirements and Research	December 2025	December 2025	Completed ✓
Design (Figma + UML)	January 2026	February 2026	Completed ✓
Sprint 1 – Auth + Setup	January 2026	February 2026	Completed ✓
Sprint 2 – Social Features	February 2026	February 2026	Completed ✓
Sprint 4 – Health + Wellness	February 2026	March 2026	Completed (extended) ✓
Sprint 6 – Accessibility	April 2026 (planned)	Ongoing concurrent	In Progress
Sprint 3 – AI Integration	March 2026	Ongoing	In Progress
Sprint 5 – Caretaker Portal	March 2026	Ongoing	In Progress
Testing and Evaluation	April 2026	April 2026	Planned

At the time of writing, Sprints 3 and 5, the AI integration via the Hugging Face Sentiment Analysis API and the caretaker portal respectively are in active development. Both are currently being debugged, which honestly is where I expected to be at this point. These are the two most technically demanding components of the system, and the fact that they are still in progress reflects a realistic understanding of the scope rather than any loss of direction. The final thesis report will document the completed and tested versions of both in full.

The current state of ElderConnect is a functioning mobile application with a fully implemented and accessibility-tested user interface, working authentication, a live social feed, health and wellness features, and the structural foundations for the AI and caretaker components already in place. That represents a substantial portion of the total system, and it was built in a way I am confident will support the remaining development cleanly.

3.2 Challenges, Mistakes and Lessons Learned

The most significant challenge I encountered during this project was not a technical one in the traditional sense, it was the challenge of designing something I could not fully evaluate from my own perspective. I am not an elderly user. I do not experience the same visual fatigue. The same hesitation before tapping an unfamiliar button, or the same anxiety about making a mistake in an application that I do not fully understand. This sounds obvious, but its practical implications ran much deeper than I initially thought.

When I built the first prototype of the ElderConnect interface, I was satisfied with it. The layout made sense, the navigation was clean, and the colour scheme felt considered. My supervisor, Ms. Dilushinie Fernando, reviewed it and recommended changing the colour scheme. Her feedback was not that it looked wrong in any conventional design sense, it was that the contrast levels and colour pairings I had chosen, while visually acceptable to a younger eye, were not reliable enough for users with the kinds of vision changes common in the target age group. I revised the scheme, ran it through contrast accessibility standards, and the difference in the subsequent usability sessions was noticeable. That one review session taught me more about designing for accessibility than the guidelines I had read during the literature review phase, because it grounded the abstract principle in a concrete and visible outcome.

The broader mistake that I did in this part was part was underestimating how much time a genuine accessibility-driven design iteration would take. I had allocated time in the plan for design, and I had allocated time for Sprint 6 as a refinement phase, but I had not fully internalised the fact that for this particular application, refinement is not a phase, it is a continuous process embedded in every build decision. Each feature I implemented raised new accessibility questions. The medication reminder notification system, for instance, required multiple iterations before the visual and audio alert combination worked in a way that was genuinely useful rather than overwhelming. The wellness games section needed the interaction patterns reconsidered after early testing revealed that the tap targets were not sufficiently sized for users with reduced motor precision. These were not catastrophic setbacks, they were exactly the kind of iterative discoveries that user testing is supposed to surface. The mistake was not expecting to encounter them; it was not having built enough time buffer into the plan to absorb them comfortably.

Another lesson I learnt from this project was about the value of structured documentation practices that I had initially viewed as administrative overhead. Creating UML diagrams during the design phase felt, at the time, like work that was taking me away from building the system. In retrospect, the use case diagram in particular proved genuinely useful during Sprint 5 development. Having a clear visual model of how the elderly user and caretaker interact with the system, what they share, and where their data flows diverge, made the caretaker portal architecture considerably easier to reason about. I would not have reached the same level of structural clarity through code alone. The progress tracking approach of monthly reports, sprint reviews, milestone sign-offs, served a similar function. By the documentation of progress I have articulated what had actually been achieved by me, rather than what had been felt to be achieved by me, and those two things are not always the same.

The AI integration in Sprint 3 has thrown up a different kind of challenge. Connecting the Hugging Face Sentiment Analysis API and building the mood scoring logic that feeds into the caretaker dashboard requires careful thinking about edge cases what happens when the sentiment result is ambiguous, how often the system should sample text, and how it presents emotional data without creating unnecessary alarm for caretakers. These are not purely technical questions and I did not want to treat them as such who the system is built for. The debugging has taken longer than I have planned, yet I would rather get it correct than rush it.

3.3 Good Practices Applied

There were a few things I have done during this project that, looking back, made more of a difference than I expected at the time. Some were deliberate decisions I had thought through carefully and others were just the right call made at the right moment, which I only appreciated why later.

Doing the primary research before compiling any code was one of those decisions that felt obvious at the time but turned out to matter more than I anticipated. When 46.2% of elderly respondents needed help completing a paper questionnaire, not an online form, a paper one that told me something important before I had written a single screen. It meant every interface decision I made from that point forward had something real behind it rather than just my own assumptions about what an elderly user might need. That grounding showed up constantly throughout development, often in small ways that are hard to pinpoint but easy to feel.

The sprint structure helped more than I expected too. I had used Agile before and understood it in theory, but working through six, two-week cycles on a single project made the value of it differently. Dedicating Sprint 1 specifically to authentication and Sprint 2 to social features established a clear definition of 'done' for each development period. The sprint review mattered more than I initially gave it credit for. Knowing that I had to show something working by a set date pushed me to prioritise differently than I would have otherwise, there is a real difference between code that runs and code that is half-written. And the deadline made sure I always had the former.

Figma ended up being useful in ways I did not fully anticipate going in. When Ms. Dilushinie Fernando reviewed the first prototype and recommended changing the colour scheme, I updated the Figma file, reviewed it, got comfortable with it, and only then touched the Flutter code. What could easily have been a multi-day re-work ended up taking about only two hours. That is the kind of saving that is easy to overlook because the problem never fully materialised. The reason it did not was precisely because the design step happened before the build step, not after.

The usability testing sessions are the thing I would push back hardest on if anyone called them informal. Yes, they were not a formal UAT phase. But running them throughout the build rather than leaving them to the end meant problems got caught early enough to actually fix within the same sprint. The tap target sizing in Sprint 4 is the clearest example, that issue was found and resolved during development, not during post-build testing. The colour contrast problem was caught at prototype stage. Neither of those required a backlog entry or a rework cycle because they were found at the right moment. That is not luck, it is what happens when testing is woven into the process rather than bolted on at the end.

When I add up what these practices actually produced: 52 elderly users and 13 caretakers responded to the surveys, which gave the requirements a credibility that would not have been there with smaller or less considered data collection. The GitHub version control meant that when an experimental navigation approach in Sprint 2 did not work, getting back to the previous state was a matter of minutes. These are not impressive-sounding numbers for their own sake, they represent time that was saved and problems that did not escalate because the right habits were in place from the beginning.

3.4 If Given a Second Chance

There are a few things I would do differently, and I think being honest about them is more useful than presenting the project as a sequence of correct decisions.

The most significant change involves the AI mood detection feature. The current design runs sentiment analysis on text posts and messages, classifies the emotional tone, and passes summaries through to the caretaker dashboard. That scope made sense on paper, but actually building it revealed how much harder emotional interpretation from text is than I had assumed especially for a user group whose writing style is often short, context-heavy, and quite different from anything the Hugging Face models were likely trained on. If I were starting again, I would treat that mismatch as a research question in its own right and build a proper evaluation of how well general-purpose sentiment APIs actually perform on elderly-generated text before committing the sprint plan to it. I also think a deliberate daily check-in prompt alongside the passive analysis would have been worth designing in from the start as something that gives the user an active role rather than having the system quietly draw conclusions from whatever they happened to write that day.

I would also have planned Sprint 6 as a concurrent track from day one rather than a standalone final phase. Running accessibility work in parallel throughout the sprints was the right call and I do not regret making it, but it meant that every progress report required some version of the same explanation that the plan said one thing and the actual workflow looked different. That is a small thing but it added up over five months of reporting.

The ethical dimension is the area I am most honest about getting wrong in terms of timing. The moment that brought me into focus of it was early in Sprint 3, when I was working out how mood data would appear on the caretaker dashboard. I realised I had not thought carefully enough about what it would feel like for an elderly user to know that a message they had written to a friend might be analysed and flagged to a family member even with consent given at registration. That concern reshaped me about how the consent model works, and I am glad it came up when it did. But it came up because I was mid-build and ran into it, not because I had planned for it. A system built for vulnerable users deserves that conversation at the planning stage, written out properly, not picked up halfway through a sprint when the architecture is already half-committed.

4. Detailed Thesis Contents

The final thesis for ElderConnect will be structured across seven chapters. The structure is outlined below, followed by a brief summary of what each chapter will contain.

Thesis Table of Contents

Preliminary Pages

Title Page | Abstract | Dedication | Acknowledgements | Table of Contents | List of Figures | List of Tables | List of Abbreviations

Chapter 1: Introduction

1.1 Project Background
1.2 Problem Statement
1.3 Aim and Objectives
1.4 Description of the Artefact
1.5 Project Scope
1.6 Structure of the Report

Chapter 2: Literature Review and Market Research

2.1 Introduction
2.2 Ageing Population and Social Isolation
2.3 Elderly Users and Technology Adoption
2.4 Accessible Interface Design for Elderly Users
2.5 Artificial Intelligence in Healthcare
2.6 Sentiment Analysis and Mood Detection
2.7 Existing Systems and Comparative Analysis
2.8 Market Research Findings
2.9 Research Gap and Conclusion

Chapter 3: Artefact Design

3.1 Development Methodology
3.2 Requirements Gathering and Analysis
3.3 System Architecture and Technology Stack
3.4 Database Design
3.5 User Interface Design
3.6 UML and Behavioural Models
3.7 Testing and Evaluation Strategy
3.8 Risk Analysis and Mitigation

Chapter 4: Development

4.1 Sprint 1 – Environment Setup and Authentication
4.2 Sprint 2 – Social Features
4.3 Sprint 3 – AI Sentiment Integration
4.4 Sprint 4 – Health and Wellness Features

4.5 Sprint 5 – Caretaker Portal
4.6 Sprint 6 – Accessibility and Refinement

Chapter 5: Testing

5.1 Testing Strategy
5.2 Unit Testing
5.3 Integration Testing
5.4 System Testing
5.5 User Acceptance and Accessibility Testing

Chapter 6: Results and Evaluation

6.1 System Results and Screenshots
6.2 Evaluation Against Objectives
6.3 Limitations

Chapter 7: Conclusion and Further Work

7.1 Project Summary
7.2 Recommendations and Future Work

References

Appendices

The preliminary pages will include a title page, an abstract of approximately 300 words, dedication, acknowledgements, table of contents, list of figures, list of tables, and a list of abbreviations.

Chapter 1 sets out the context for the project, opening with the global ageing population challenge and narrowing to the specific gap that what the ElderConnect addresses. It presents the problem statement, aim, and seven measurable objectives, then describes both portals, the Elderly User Portal and the Caretaker Portal alongside the project scope and report structure.

Chapter 2 builds the theoretical and empirical foundation. The literature review spans five areas: social isolation in elderly populations, barriers to technology adoption, accessible design principles, AI in healthcare, and sentiment analysis techniques. Survey findings from 52 elderly users and 13 caretakers are then presented, followed by a comparative analysis of existing systems and a clear articulation of the research gap.

Chapter 3 covers every design decision made prior to and during development - methodology justification, requirements analysis, system architecture, and the rationale for selecting Supabase over the initially proposed MongoDB given the relational nature of the caretaker-user data model. The normalised database schema, Figma prototypes, UML diagrams, testing strategy, and risk table are all documented here.

Chapter 4 walks through the build sprint by sprint, from the Supabase environment setup and dual-role authentication in Sprint 1 through to the accessibility refinements in Sprint 6. Each sprint section is supported by code segments, implementation decisions, and annotated screenshots as technical evidence.

Chapter 5 covers the full testing lifecycle; unit, integration, system, and user acceptance testings. Structured test case tables present expected and actual results for each feature, and the accessibility testing subsection documents evaluation conducted specifically with target-age participants.

Chapter 6 presents the completed system through screenshots and feature demonstrations, then evaluates the project against each of its seven objectives. Limitations are discussed honestly, particularly regarding the Hugging Face API's performance on elderly-generated text, with user acceptance feedback incorporated throughout.

Chapter 7 summarises the project achievements and reflects on delivery against the original plan. The further work section proposes future enhancements as, multi-language support, voice-based sentiment analysis, video calling, and potential integration with external healthcare records, positioning the current version a strong foundation for continued development.

References are compiled in full Harvard format. Appendices include the survey questionnaires, Figma wireframes, detailed test case tables, and supporting evidence from the development and testing phases.

Appendices

Appendix A — Updated Gantt Chart

Figure 1 shows the original Gantt chart submitted with the contextual report in December 2025. Figure 2 shows the updated version reflecting actual progression as of March 2026. Comparing the two illustrates the key deviations discussed in Section 3.1: the extension of Sprint 4 due to iterative UI refinements driven by usability testing, the reclassification of Sprint 6 from a final standalone phase into a concurrent accessibility track running throughout development, and the extension of Sprints 3 and 5 beyond their originally planned dates as debugging continues. The addition of informal usability testing sessions which is not present in the original plan is also visible in the updated chart as a recurring activity across multiple sprints.

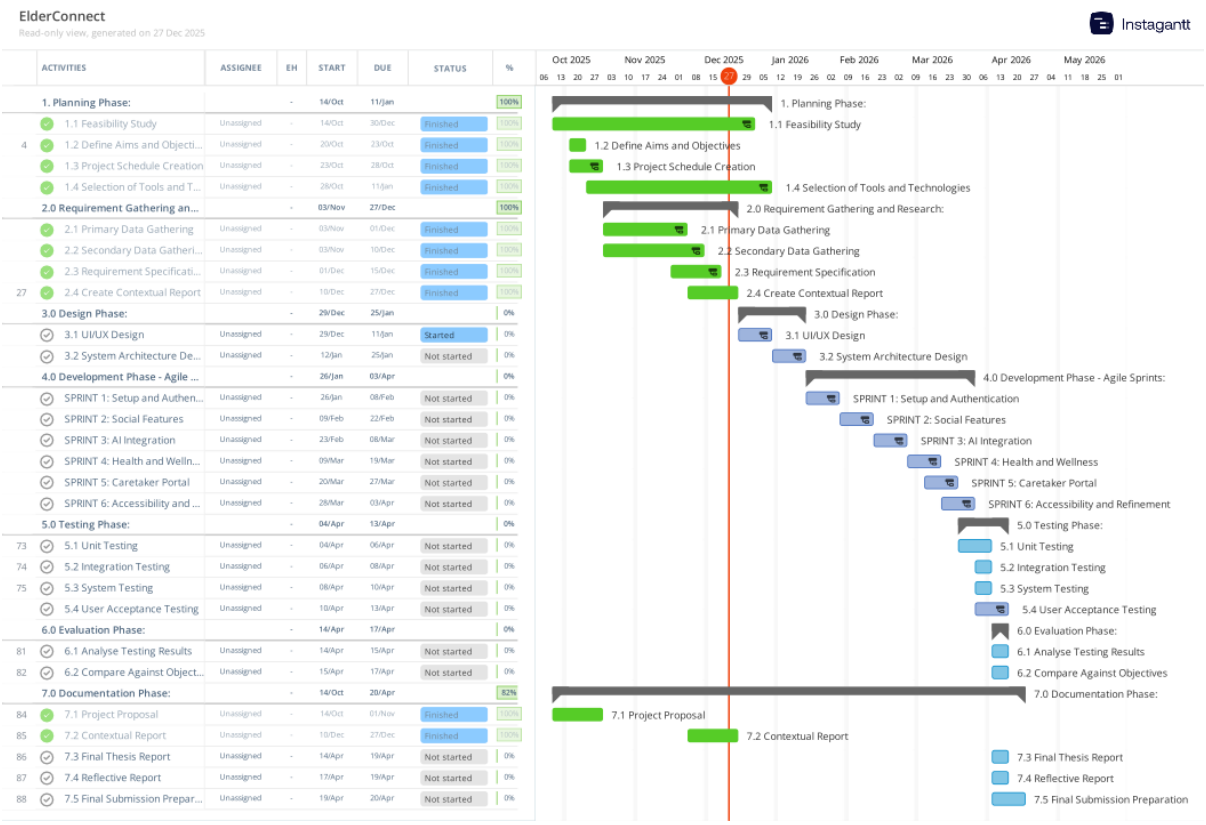

Figure 1: Original Gantt Chart

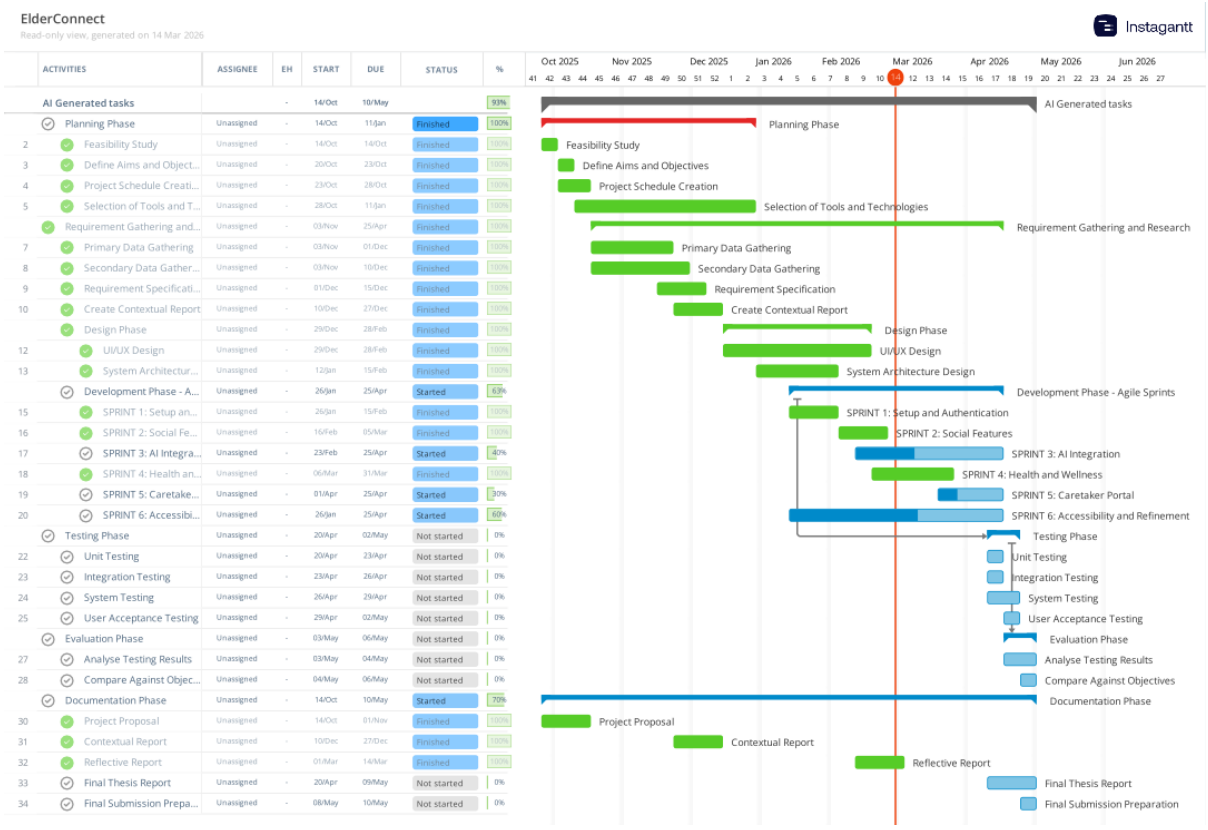

Figure 2: Updated Gantt Chart

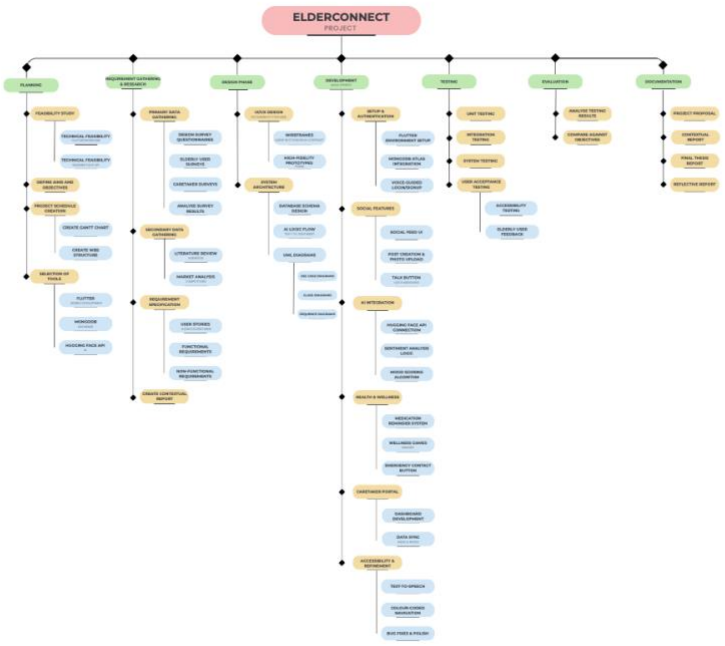

Figure 3: WBS for ElderConnect

Appendix B — Project Screenshots

The following screenshots provide visual evidence of the current state of the ElderConnect application. They demonstrate the accessibility-first design approach applied across the implemented screens, including the colour-coded navigation system, large-button interface layout, and the social feed. These screens were developed through the iterative design and usability testing process described in Section 3.1 and Section 3.2 of this report.

Figure 4: Home Page - Elder

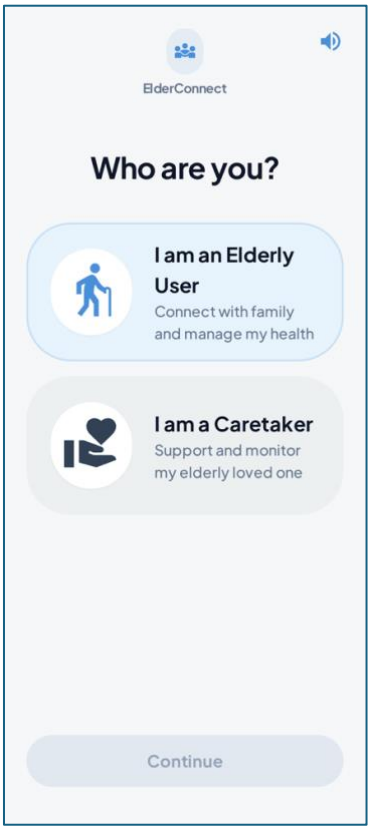

Figure 5: Dashboard Selector

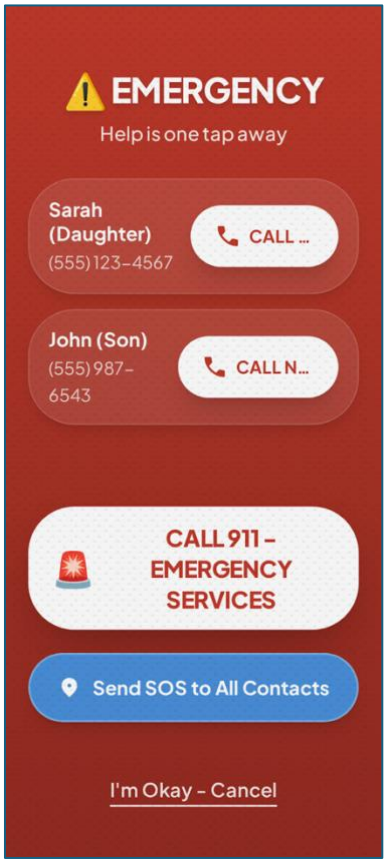

Figure 6: Emergency Contact

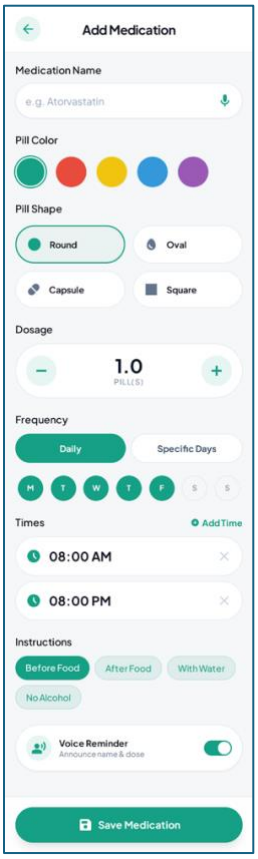

Figure 7: Add Medication

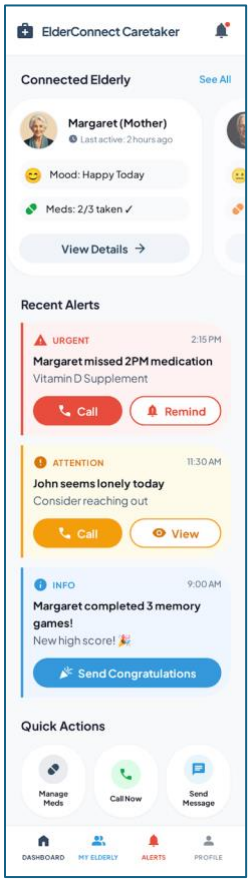

Figure 8: Caretaker Home Page